

Review

Artificial intelligence and infectious diseases: Scope and perspectives

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ABSTRACT

Artificial intelligence (AI) is set to permeate every facet of infectious disease practice—from prevention and public health surveillance to epidemic management and bedside care. Routine care data (laboratory results, medication orders, progress notes) and research-generated datasets now fuel state-of-the-art machine-learning (ML) pipelines that sharpen diagnosis, prognosis, antimicrobial stewardship, and, by combining both sources, accelerate drug discovery. In diagnostics, deep networks that now flag pneumonia or tuberculosis on chest images are increasingly able to identify—and localize—virtually more infectious processes throughout the body, while simultaneously predicting pathogen identity and antimicrobial resistance from routine microbiology. Prognostic models trained on Electronic Health Records surpass traditional scores in anticipating clinical deterioration or postoperative sepsis, enabling earlier targeted interventions. Predictive analytics can also personalize antimicrobial dosing by fusing real-time drug-monitoring data. Large language models (LLMs) build upon these advances by transforming unstructured clinical narratives into structured phenotypes suitable for predictive modeling, automatically summarizing patient encounters, generating synthetic cohorts for rare conditions, and providing real-time conversational decision support at the patient's bedside. Despite rapid progress, real-world deployment faces hurdles: high computational and licensing costs, vendor-specific implementation constraints, limited cross-site model transferability, and fragmented governance of safety, bias, and cybersecurity risks. Rigorous, lifecycle-based evaluation frameworks—covering external validation, cost-effectiveness analysis, and post-deployment monitoring—are required to ensure safe, equitable, and sustainable AI adoption. This review synthesizes current applications, evidential strengths, and unresolved challenges, and proposes a translational roadmap aligning technical innovation with clinical and regulatory realities.

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1. Introduction

1.1. Background

Artificial intelligence (AI) is now part of our daily lives, and its role in health care continues to expand. In the field of infectious diseases, AI is already used in various applications, such as supporting microbiological diagnosis [1] or forecasting epidemics [2]. However, the full potential of AI in infectious diseases has yet to be fully leveraged. Infectious diseases present unique and complex challenges—including the emergence of pathogens, epidemic dynamics, the complexity of clinical trajectories, and the interconnectedness with environmental and animal health—that are particularly well-suited to AI integration [3]. AI tools offer the capacity to rapidly process large volumes of heterogeneous data to support clinical decision-making, to guide public health strategies, and to detect weak signals at the population level. In this article, we discuss the main contributions of AI in infectious diseases to date, future development perspectives, and the challenges that remain to be addressed (Fig. 1).

1.2. Basic concepts

AI, understood as the set of techniques that allow a machine to simulate certain human cognitive functions, has its origins as early as the 1950s. The first AI systems, known as *symbolic*, rely on the explicit encoding of logical rules derived from human expertise, as seen in expert systems. These pioneering approaches quickly showed their limits when faced with the growing complexity and volume of knowledge to be integrated, as well as their lack of flexibility. Since the 2010s, AI has undergone major revival with the progress in *machine learning*, supported

by advances in computing and the availability of large and diverse datasets via the internet [4]. In machine learning, the model learns directly from data that is either labeled (e.g., images of bacteria annotated as cocci or bacilli) – known as supervised learning – or unlabeled (e.g., bacterial images without annotation) – referred to as unsupervised learning. Deep learning is a subcategory of machine learning that relies on neural networks. These networks enable the construction of complex representations of data through multiple layers of processing [5]. Machine learning models are used in a wide range of applications, such as prediction or classification tasks. In recent years, the rapid emergence of generative artificial intelligence models—particularly large language models (LLMs) such as ChatGPT—has marked a major turning point in the evolution of AI. Unlike earlier machine learning systems, which were often costly to train and narrowly designed for a single, predefined task (such as diagnosing pneumonia from chest X-rays or classifying bacterial morphologies), these new models are trained to predict the next word in a sequence across vast corpora of general or domain-specific texts. Through this seemingly simple objective, they acquire an emergent capacity to solve a near-infinite variety of tasks, many of which were not explicitly anticipated during training. This shift, from task-specific, hard-coded classifiers to general-purpose, instruction-following models, represents a profound rupture in the design and deployment of AI in medicine. It opens the door to applications as diverse as automated clinical summarization, reasoning over patient trajectories, generating synthetic data, or providing conversational decision support. However, this new paradigm also introduces novel challenges, particularly in ensuring reliability, explainability, and clinical safety, as these models can produce plausible yet inaccurate outputs—so-called “hallucinations”—in high-stake contexts [6].

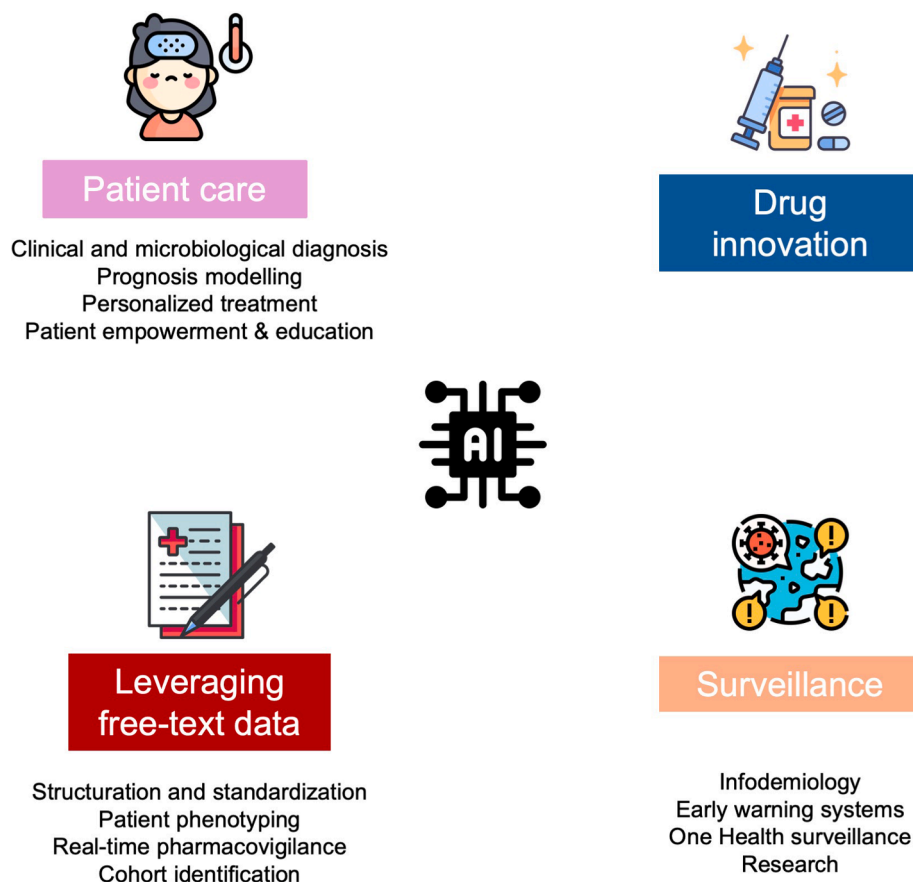


Fig. 1. Artificial intelligence in infectious diseases: patient care, drug innovation, and public health surveillance.

2. AI and infectious diseases

2.1. AI for diagnostics support in infectious diseases

AI is increasingly augmenting diagnostic workflows for infectious diseases, especially through advanced pattern recognition in medical images and laboratory data. In medical imaging, AI models can detect infections like pneumonia or tuberculosis (TB) on chest radiographs and CT scans with accuracy approaching expert radiologists. For example, in COVID-19 pneumonia, deep learning systems have achieved sensitivities of 60–95 % on test datasets, comparable to human radiologists (range 42–100 %) [1]. Notably, combining AI with physician readings can improve consistency and boost sensitivity into the 81–98 % range [7]. AI tools have also shown the ability to distinguish COVID-19 from other pneumonias on imaging, potentially enabling faster triage and isolation decisions. In TB screening, several commercial AI algorithms met the World Health Organization's targets for an effective triage test, reaching ≥ 90 % sensitivity with 73–74 % specificity [8]. At this high-sensitivity operating point, these AI systems even outperformed human readers in specificity, reducing false positives and halving the number of confirmatory GeneXpert tests needed. Such findings underscore that AI can match or exceed traditional diagnostic methods in detecting infectious diseases on imaging, translating to practical benefits like fewer unnecessary tests and more consistent interpretation across readers.

Beyond radiology, AI-driven diagnostics have expanded into microbiology laboratories. Machine learning models can analyze complex laboratory data – for instance, mass spectrometry profiles – to identify pathogens and predict antimicrobial resistance. A recent large-scale study applied AI to $>300,000$ MALDI-TOF mass spectra from clinical isolates and successfully predicted antibiotic resistance for key pathogens (e.g., *S. aureus*, *E. coli*), with area under the ROC curve (AUROC) values of 0.74–0.80 [9]. This approach could deliver susceptibility insights 1–2 days faster than traditional culture-based methods, which often take approximately 72 h. In a retrospective analysis, the MALDI-AI method would have prompted earlier effective therapy in 14 % of cases, with 89 % of those treatment changes benefiting patients [9]. Other AI applications under exploration include automated classification of digitized microscope images (for example, malaria blood smears [10] or Gram stains) and rapid genomic analysis to detect resistance genes [11]. While these are still emerging, they illustrate the broad diagnostic reach of AI – from imaging suites to bench-top analyzers – in accelerating pathogen detection and supporting clinicians in early, accurate diagnosis.

2.2. AI for prognostic modelling

AI is also transforming prognostic modelling in infectious diseases, enabling a shift from reactive care to proactive, data-driven decision-making. Machine learning models can integrate real-world clinical and biomarker data to predict patient trajectories – for example, identifying which patients with infection are at high risk of deteriorating, developing sepsis, or requiring intensive care unit (ICU) transfer. These predictive models often outperform traditional risk scores in sensitivity and timeliness. In multiple ICU studies, AI-based early warning systems for sepsis have achieved AUROCs in the 0.85–0.95 range [12], significantly higher than conventional scoring tools like qSOFA or MEWS (which typically show AUROCs 0.70–0.80). In one example, a gradient boosting model attained an AUC of approximately 0.89 for impending sepsis, substantially outperforming the SOFA score's accuracy [6]. Importantly, these models can detect clinical deterioration hours in advance, giving clinicians a valuable head-start. For instance, an AI might flag a ward patient as high-risk for septic shock, prompting closer monitoring or preemptive ICU admission before organ failure sets in.

Real-world impact of AI prognostic tools is now being documented. Not all models have succeeded – a cautionary example is the widely used Epic Sepsis Model, which an external study found to have poor

calibration (AUROC 0.63) and low sensitivity (approximately 33 % at 83 % specificity) in practice [13]. This model missed two-thirds of sepsis cases while triggering frequent false alerts, highlighting the importance of rigorous validation and oversight. In contrast, newer AI-driven systems have demonstrated clear clinical benefits when thoughtfully implemented. A recent quasi-experimental study at two large hospitals evaluated an AI early warning system for sepsis (the “COMPOSER” model) and observed a 17 % relative reduction in sepsis mortality after deployment (absolute mortality drop around 1.9 %) [14]. Use of the AI alerts was also associated with a 5 % absolute increase in compliance with the sepsis treatment bundle and slower progression of organ failure, indicating that clinicians intervened earlier and more aggressively when prompted. Likewise, prognostic models for COVID-19 have aided in therapeutic stratification: adding an AI analysis of initial chest X-rays to clinical data significantly improved prediction of which patients would progress to critical illness (external-validation AUROC improved from 0.73 to 0.79) [15]. Such tools allow care teams to anticipate complications – for example, by identifying a COVID-19 patient likely to require ventilation or a bacteremic patient at risk of septic shock – and to initiate tailored treatments or monitoring protocols proactively.

AI has also been used to predict postoperative infections, which are among the most common and costly nosocomial complications affecting up to 18 % of surgical patients. In the PERISCOPE study, XGBoost models showed high performances in predicting postoperative infection within 7 and 30 days after surgery in three hospitals, leveraging automatically extracted data from the electronic health record (HER). Developed models also showed higher clinical utility as compared to the ‘treat all’ or ‘treat none’ approaches. The models were externally validated on temporal validation datasets and locally updated in two of the hospitals after an initial training on the data from the first hospital. Locally updated models had better performances and reached higher clinical utility. These results highlight that local validation and updating of AI models may be an interesting strategy to facilitate their deployment in clinical practice.

Overall, AI-based prognostic modeling is steering infectious disease management toward earlier intervention and personalization. By forecasting risks like sepsis, respiratory failure, or adverse drug reactions, these systems support clinicians in making informed decisions (e.g., escalating care, adjusting antimicrobial therapy) before a crisis unfolds. This proactive approach, empowered by large EHR datasets and advanced algorithms, has begun to show improved clinical outcomes in studies. While challenges remain regarding data quality, bias, and user trust, the evidence to date demonstrates that AI can enhance both diagnostic precision and therapeutic decision-making – complementing clinical judgment with robust predictions to ultimately improve patient care in infectious diseases.

2.3. AI in antimicrobial stewardship: From policy to practice

The fight against antimicrobial resistance (AMR) calls not only for better diagnostics and novel antibiotics, but also for optimized use of existing treatments. Antimicrobial stewardship (AMS) programs aim to ensure that antibiotics are used rationally, adapted to individual patient needs, and aligned with ecological and public health priorities. AI, particularly through machine learning (ML), is now contributing significantly to this mission. Recent advances show that AI can support AMS at multiple levels – from personalizing antibiotic dosing, to guiding empirical therapy, to real-time surveillance and feedback. Two complementary reviews [16,17] offer a comprehensive view of how AI is reshaping stewardship practice in infectious diseases.

These innovations serve a common goal: reducing unnecessary antibiotic exposure, improving clinical outcomes, and curbing the emergence of resistant pathogens.

2.3.1. AMS and resistance mitigation

AI has emerged as a key enabler of modern AMS efforts by enhancing

the precision, timeliness, and adaptability of antibiotic use. In a recent systematic review and *meta-analysis* encompassing 80 studies and over 1.3 million patients, machine learning (ML) models demonstrated robust performance in supporting AMS decisions across clinical settings. The studies reviewed covered diverse applications of ML in AMS, including predicting resistance patterns, assessing the appropriateness of empirical treatments, and identifying patients suitable for early intravenous-to-oral switch. The pooled predictive metrics were notably strong: AUC 72.3 %, accuracy 75.0 %, sensitivity 76.9 %, specificity 73.8 %, NPV 79.9 %, and PPV 69.4 % [17]. The majority of models used Random Forest (RF) or gradient boosting algorithms such as XGBoost (XGB), both of which demonstrated consistently high discriminative ability across infection types—namely urinary tract infections (UTIs), bloodstream infections, and respiratory infections [17]. Some AI systems were trained directly on microbiological data of various nature (MALDI-TOF mass spectra, bacterial ecology data, etc.) to anticipate antibiotic susceptibility before traditional culture results became available, thereby accelerating the initiation of effective, narrow-spectrum therapy [18]. These results support the integration of AI tools in real-time antibiotic decision-making to reduce inappropriate prescribing and limit the spread of resistance.

Importantly, while internal validation was common, external validation was scarce (only 7 of 80 studies), limiting generalizability and highlighting the need for prospective implementation studies in diverse healthcare environments [17]. Nonetheless, AI-based models have already shown their potential to inform prescribing in complex scenarios. For instance, predictive models for extended-spectrum beta-lactamase (ESBL) production enabled stratification of patients likely to benefit from narrower-spectrum agents in endemic settings, reducing unnecessary use of carbapenems. Similarly, ML systems applied to real-world hospital datasets allowed for accurate identification of multidrug-resistant organisms (MDROs), guiding early isolation decisions and tailoring of empirical therapy.

These tools also support stewardship teams in identifying outliers and benchmarking performance. Some systems automatically flag potentially inappropriate prescriptions or deviations from local guidelines, allowing for timely feedback to prescribers. Others can continuously adapt to evolving resistance trends by retraining on updated EHR and microbiology data, creating a dynamic, learning stewardship infrastructure.

2.4. AI for therapeutic innovation

Virtual screening, also known as *in silico* screening, is a computational method used in pharmaceutical research and structural biology to test millions of chemical compounds. This technique helps identify those that may have promising biological activity against a specific target, such as a protein involved in a disease, including viral, parasitic, or bacterial infections. By drastically reducing the number of compounds that need to be tested experimentally, virtual screening saves both time and resources. These so-called computational methods involve the use of computer-based tools and techniques to solve complex problems in science and engineering. They rely on mathematical models, algorithms, and simulations to analyze and predict the behavior of physical, biological, and chemical systems [19]. Thanks to deep learning, it has become possible to assess the antimalarial activity and cytotoxicity of various chemical compounds. Two molecules, LabMol-149 and LabMol-152, have shown particularly promising results [20].

Molecular docking is a technique used in virtual screening to predict how a ligand will bind to a biological target. This method helps evaluate the affinity and stability of molecular interactions. For example, a study analyzing the binding affinity of 33 compounds with inhibitory activity against the HIV-1 protease led to the development of a model capable of identifying a new molecule with activity against HIV-1 protease. Ultimately, its effectiveness was found to be comparable to darunavir [21]. In the context of emerging antimicrobial resistance, it is easy to see how

these new techniques are invaluable for the development of novel antiviral, antibacterial, antifungal, or antiparasitic agents [21].

2.5. AI to support epidemics detection and surveillance

The integration of AI into epidemiology offers promising prospects for epidemic detection and surveillance. These technologies can be leveraged in various ways to improve public health [22].

Infodemiology, or information epidemiology, is a subfield of health informatics defined as “the science of the distribution and determinants of information in an electronic medium, specifically the Internet, or in a population, with the ultimate aim to inform public health and public policy” [23]. By analyzing search queries on search engines, infodemiology enables the study of epidemics. For instance, a study using Google Trends (a free online tool that tracks the popularity of search terms over time) demonstrated how the terms “hotel”, “poor hygiene”, “blueberries”, and “hypovolemic shock” correlated with the epidemiology of salmonellosis [24].

Early warning systems are also predictive tools that use various data sources. They are designed to detect and signal imminent risks early on, allowing for timely preventive actions. These systems are applied across different domains, including epidemics and natural disasters such as cyclones or floods. For example, the ARNN-LE (Auto-Reservoir Neural Network), an AI-based method using neural networks fed with temporal and regional data, was used to implement an early warning system for COVID-19 spread across six different countries [25]. The developed model helped anticipate public health interventions such as SARS-CoV-2 vaccination campaigns or population-wide isolation measures.

Time-series models can also be used to forecast outbreaks linked to climate change. Patil *et al.* developed a predictive model for dengue incidence using 10 years of data from nine cities in the state of Maharashtra, India. Model inputs included minimum and maximum temperatures, humidity, rainfall, wind, and monthly dengue incidence. This model helps anticipate necessary preventive actions in the face of climate change and supports tailored health policies for dengue outbreaks, such as vaccine research support, vector control, or public education campaigns [26].

AI can also be used to optimize health care organization during epidemic situations. Hinson *et al.* developed a machine learning algorithm capable of predicting intensive care needs within 24 h and general hospitalization needs within 72 h during a COVID-19 outbreak. This tool was rapidly implemented in clinical practice in five emergency departments across the United States, contributing to more consistent and reliable decision-making and better bed allocation during the pandemic [27].

Surveillance systems around the world, which have traditionally relied on standardized human laboratory data, are now progressively incorporating additional data sources—including clinical, animal, and environmental data—within a One Health approach [28]. With the rapid development of clinical data warehouses and data sharing platforms on national and international levels, increasing efforts are being made to integrate real-world data into surveillance systems, moving toward real-time epidemiological monitoring [29]. Advances in genomics are also opening new horizons by enabling the integration of genetic data that enhance our understanding of the mechanisms underlying the development and spread of infectious agents and our capacity of detecting outbreaks among humans, animals and the environment [30,31]. Through its ability to rapidly and effectively analyze such complex, multimodal data, AI expands the possibilities for transforming surveillance [32]. It enables the integration and interpretation of diverse data sources to support the development of faster, more responsive surveillance systems, better equipped to meet tomorrow's infectious disease challenges.

2.6. Natural language processing and large language models for the analysis of electronic health records in clinical care and research

The application of AI to EHR has opened a new frontier for both clinical care and medical research, particularly in infectious diseases. Unstructured clinical notes—long considered a rich yet inaccessible source of information—can now be mined effectively through natural language processing (NLP) and LLMs. These tools are transforming how clinicians and researchers extract, structure, and utilize health information, enabling advances in real-world evidence generation, patient stratification, adverse event surveillance, and individualized decision-making.

2.6.1. From free text to structured knowledge

Traditionally, the wealth of clinical nuance documented in free-text narratives (e.g., progress notes, discharge summaries, consultation letters) was largely untapped due to its unstructured nature. Yet these notes often contain vital data—symptoms, temporal dynamics, contextual factors, and diagnostic hypotheses—that are absent from structured fields. NLP addresses this challenge by parsing human language and mapping it to standardized clinical concepts. More recent advances with LLMs such as BioClinicalBERT, GatorTron, and GPT-derived models have substantially improved performance in information extraction, medical entity recognition, and temporal reasoning [33].

These models enable large-scale transformation of narrative content into structured variables. For instance, Yuan *et al.* [34] developed a semi-supervised NLP pipeline to extract tumor stage, histological subtype, and gene mutations from unstructured cancer records, leading to significantly improved survival predictions. Similar methods in infectious diseases can extract antibiotic regimens, timing of symptom onset, or microbiological results—key variables for evaluating care quality or infection dynamics.

2.6.2. Enabling patient phenotyping and cohort identification

A key benefit of NLP is the ability to retrospectively identify clinically relevant cohorts for research and care optimization. Rather than relying solely on ICD codes—which are often incomplete—NLP pipelines can detect complex phenotypes based on patterns in physician notes, microbiology results, and treatment trajectories. GatorTron, a transformer-based LLM trained on over 90 billion words of de-identified clinical text, demonstrated state-of-the-art performance in concept extraction and inference tasks across multiple benchmarks, outperforming smaller models by over 10 % on medical question answering and entailment [33,35].

These advances facilitate the creation of high-resolution patient datasets for infectious disease studies, such as for COVID-19, HIV/AIDS, or multidrug-resistant infections. By retrieving contextual information such as immune status, exposure history, or prior antimicrobial response, NLP can enable fine-grained phenotyping for observational studies and pragmatic trials. In the study of Yuan *et al.* (2021), NLP-extracted features enabled improved prognostic modelling in lung cancer, with AUROC increasing from 0.88 to 0.93 compared to models using structured data alone [34].

2.6.3. Surveillance of adverse events and pharmacovigilance

Free-text documentation is often the only place where adverse drug events (ADEs) or complications are recorded. NLP and deep learning have been shown to enhance pharmacovigilance by identifying these mentions and linking them to specific therapies. In a 2025 scoping review, Golder *et al.* reported that NLP models consistently detected ADEs missed by structured EHR fields, improving hospital safety monitoring systems [36]. The review synthesizes over 70 studies, including several that used free-text mining to uncover mild to severe ADEs, including non-coded reactions and events that would have been missed in conventional chart abstraction, such as gastrointestinal bleeding linked to certain antibiotic regimens or “nausea after piperacillin-tazobactam”.

NLP models trained to read discharge summaries and clinical notes were able to identify unreported reactions, such as “nausea after piperacillin-tazobactam”, often absent from formal allergy lists or pharmacovigilance logs. This enhanced capacity to monitor adverse outcomes is particularly relevant in infectious disease contexts where off-label, empirical, or toxic treatments are frequent, such as in sepsis or invasive fungal infections.

2.6.4. Similar patient retrieval and decision support

LLMs also open the possibility of retrieving “similar patients” from hospital data to support clinicians facing uncertain or complex cases. Instead of relying solely on structured matching (e.g., diagnosis codes), AI models can compare entire narrative histories and identify patients with overlapping patterns of evolution, immunologic context, or response to treatment. This method, sometimes called patient-level embedding retrieval, has been proposed to support personalized treatment in oncology, critical care, and increasingly, infectious diseases.

For example, in the context of early sepsis detection, Shashikumar *et al.* (2025) developed COMPOSER-LLM, a hybrid system combining structured-data algorithms with an LLM that analyzes clinicians’ notes in real time [37]. The model successfully increased both sensitivity and positive predictive value for sepsis alerts by incorporating textual elements. In prospective testing across 2,500 patients, the system achieved 72 % sensitivity and halved false alarm rates compared to baseline models. This paradigm is particularly promising for complex infectious diseases where decision-making relies on a mix of microbiology, host status, empirical treatments, and evolving clinical signs—elements best captured in free text.

2.6.5. Enhancing data for research and population health

NLP also enriches EHR data for secondary use in research and public health. A critical example is the extraction of social determinants of health (SDoH), often mentioned only in notes. Guevara *et al.* fine-tuned T5-based models to identify SDoH categories such as housing instability, employment concerns, or lack of insurance from clinical notes [38]. Their best models captured SDoH in over 90 % of cases where they were present in text—compared to just 2 % coverage using ICD-10 codes. Moreover, these models proved less sensitive to demographic bias than general-purpose LLMs like ChatGPT. This is especially relevant in infectious diseases, where SDoH significantly modulate exposure risk, treatment adherence, and outcomes, notably for TB, HIV, or hepatitis C. As LLMs mature, their capacity to generate synthetic datasets, summarize longitudinal trajectories, or translate clinical language into patient-friendly formats may also reshape communication, training, and patient engagement. This reinforces their value beyond analytics, toward enhancing transparency and interaction in the care process.

2.7. AI for patient empowerment

Traditionally, medical information delivery has predominantly flowed from the healthcare professional to the patient. This dynamic has been significantly reshaped by internet-based communication platforms, which have facilitated the exchange of information among patients and other actors [39]. The widespread adoption of generative AI tools, such as ChatGPT, marks another turning point in this relationship. These tools, capable of answering a wide range of medical queries in natural language, are increasingly being used by patients to seek medical information. They excel at simplifying complex information and translating it into more accessible and often more empathetic language. They can also assist patients in synthesizing and translating information and help them acquire and interact with medical knowledge [40,41]. However, generative AI systems are not trained to answer based on medically validated ground truths, but to generate responses based on probabilistic associations in their training data. As a result, they may present incorrect or misleading information with high confidence—a phenomenon known as “hallucination” [42–45]. Some patients are also

poorly informed about the privacy issues associated with these tools, particularly the potential for their conversations with the model to be reused to improve its performance. Healthcare professionals need to be particularly aware of these advantages and limitations to maximize the benefits of using these tools while minimizing risks by engaging with patients, informing them, and guiding them in interpreting the information they receive.

Beyond information delivery, AI tools could be integrated in the near future into various numeric tools used for interaction between the patient and healthcare professionals. For instance, they could facilitate the follow-up of individuals receiving long-term antibiotic therapy, by interacting with the patient, collecting self-reported symptoms in natural language, automatically structuring them, and transmitting the information to a healthcare team—potentially with alert functionalities. Similarly, they could be used to monitor and document adverse drug reactions in real-time, improving pharmacovigilance in ambulatory settings.

3. Perspectives and challenges

As AI becomes increasingly integrated into healthcare, infectious disease specialists must understand not only its potential but also the technical, organizational, and ethical challenges it raises. Successful implementation requires more than performant algorithms—it depends on data quality, clinician trust, patient privacy, appropriate infrastructure, and a regulatory environment aligned with clinical realities.

3.1. Data interoperability

AI models need access to large, diverse, and well-structured data to function effectively. However, hospital data are typically fragmented across different departments (laboratory, radiology, microbiology, pharmacy) and stored in incompatible formats. This lack of interoperability prevents the development of comprehensive AI tools capable of learning from all available patient information. To address this, healthcare systems are moving toward the adoption of international data standards such as HL7, FHIR (Fast Healthcare Interoperability Resources). FHIR allows clinical data (test results, medications, diagnoses) to be represented and shared consistently across systems. Several national initiatives in Europe and North America now recommend or require its use. In parallel, terminologies such as SNOMED CT and LOINC help standardize medical concepts.

Hospitals must also improve the structure of their own clinical information systems. Many EHRs still rely heavily on free text, which is difficult to analyze. AI can help here—for instance, by suggesting structured data entry based on what the clinician writes—but system design needs to support this transformation. Ultimately, better data structure benefits not only AI, but also clinical communication, quality monitoring, and research.

3.2. Model transparency and clinical trust

AI models, especially those based on deep learning, often work in ways that are not directly understandable to humans. For infectious disease physicians, this can be a barrier: how can one trust a tool that suggests an antibiotic change or flags a patient as high risk if the reasoning is opaque?

This has led to the development of explainable AI (XAI) approaches. Some tools highlight which clinical features or document sections influence the model's output (e.g., using heatmaps or feature rankings). Others retrieve similar past cases to help clinicians understand the basis of the prediction. While these methods help, they are not perfect, and many clinicians still prefer models that are simpler but more transparent. That's why explainability must be combined with rigorous clinical validation. Even if a model cannot explain its decisions in plain language, it must prove its usefulness through prospective trials or real-

world evaluation. Without this, clinician adoption—and patient safety—are at risk.

3.3. Bias and equity

Medical AI systems learn from past data—and these data often reflect real-world inequalities. For example, if a hospital treats more men than women for endocarditis, an AI trained on its records may underdiagnose women. Similarly, certain populations may be underrepresented in microbiology datasets, leading to poorer performance of diagnostic tools. Bias can come from many sources: imbalanced datasets, differences in documentation practices, or inappropriate use of surrogate variables (e.g., using healthcare costs as a proxy for illness severity). In infectious diseases, these biases could affect decisions such as ICU triage, isolation protocols, or antibiotic selection.

To reduce these risks, researchers now routinely evaluate AI models across demographic groups. Strategies such as rebalancing data, adjusting predictions, and engaging patients during model design are becoming standard. In 2024, the “STANDING Together” consortium published practical recommendations to document data representativeness and ensure inclusive model development.

3.4. Data privacy and use of large language models

Many recent AI tools—especially LLMs like ChatGPT or specialized clinical equivalents—rely on processing free-text clinical notes. These notes often contain sensitive patient information, and even when identifiers are removed, there is still a risk that a patient could be re-identified (e.g., based on a rare condition or specific details).

This raises two critical concerns depending on how the model is used:

- Local deployment (running the model entirely within the hospital's IT infrastructure) protects data privacy but requires significant computing power, energy, and skilled personnel. Hosting an LLM locally involves high-performance servers, secure storage, software maintenance, and energy costs—far beyond the typical resources available in public hospitals.
- External services via APIs (application programming interfaces, which allow different software systems to communicate—for example, by sending data to a cloud-based model) reduce technical complexity but raise serious data protection issues.
- Even with contracts and encryption, sending patient notes to an external server—possibly outside the European Union—poses legal and ethical risks, especially under General Data Protection Regulation (GDPR).

This situation creates a paradox: AI tools are seen as promising for analyzing EHRs, summarizing patient histories, or detecting complications, but hospitals often lack the resources or legal clarity to use them safely.

Moreover, none of the costs associated with AI infrastructure—hardware, software licenses, energy, or compliance—are currently included in hospital budgets. This means clinical teams may want to use AI tools but have no funding pathway to acquire or maintain them.

3.5. Regulation and ethical frameworks

In Europe, the GDPR already sets strict rules for using personal health data. In 2025, the AI Act will introduce new obligations, especially for “high-risk” systems such as those used in clinical decision-making. These will include transparency, human oversight, post-market monitoring, and proof that the AI system performs safely and fairly.

Hospitals that use AI—whether for decision support or research—will need to ensure their tools are properly classified, documented, and monitored. Participating in regulatory sandboxes, as

proposed by data protection authorities (e.g., CNIL in France), offers a practical way to test AI tools under supervision before full deployment.

Finally, the rapid rise of LLMs poses unique ethical questions. These models can produce plausible but incorrect answers (“hallucinations”), reuse sensitive training data, or be repurposed for misinformation. In medicine, these risks are amplified. Strict governance, clear labeling of AI-generated outputs, and institutional validation will be essential to protect patients and preserve trust.

4. Conclusion

Artificial intelligence is no longer a distant promise for infectious disease specialists; it is already reshaping how we diagnose, treat, and prevent infections. From rapid genome-based identification of pathogens to bedside dosing algorithms that minimize toxicity and resistance, AI has shown that it can turn the discipline’s data deluge into actionable insight at the speed required by evolving microbes and globalized outbreaks.

Yet real-world impact demands the same scientific rigor we apply to new antimicrobials or vaccines. Prospective multicenter trials, continuous post-deployment surveillance, and transparent model reporting across diverse populations must become the norm, not the exception. Regulatory frameworks such as the EU AI Act provide a scaffold, but clinicians, data scientists, and software engineers share the responsibility to translate those principles into auditable pipelines, interpretable interfaces, and up-datable models that remain clinically relevant as epidemiology shifts.

Finally, unlocking the full potential of AI will hinge on building critical and constructive capacity within the infectious disease community itself. This means embedding data-science literacy in medical curricula, fostering “bilingual” professionals who can question algorithms as deftly as they read blood cultures, and establishing interdisciplinary teams where domain knowledge and technical innovation iterate rapidly. By cultivating such expertise—and insisting on evidence of real-world benefit—we can ensure that AI amplifies, rather than replaces, the clinical judgment and public health vision at the heart of our field.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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